①

7030* (K9) (Dog) T. C. Chen Sept 20/1961

1. 3 accumulators
2. VFL high speed (X2 for character processing)
3. index instrns (Direct 2X as fast -- all post-gressive indexing
   dramatic improvement

```
eg. → LMR(P, 1($1)          ie, put  X+1 → X
    |  *+(P), N($2)            addr field + C_v($X) → C_v($X)
    └ BZXCZ
```

LX is now 5.7 μs $\boxed{\text{1.8 μs not overlappable}}$
4.8 μs in K-2
in * (0 to 4.8 max) overlap.

4. Easy to program-well
5. Incorporates unusual debugging features (where from in unusual branch)
6. Possesses adv. interrupt + MCP features -- ("drain dry").
   - look at 5 bits

EQK
ATT
INEX
REX
DF

MCP

Prob P.

7. Multiprogrammed Machine
8. Allows a further 2X speed
9. Present estimate 1.5X for large probs
   2X for small
   2X for MTRC
10. Projected cost within 15%

Compatibility?

- instr. set completely revamped

I/O std. interface

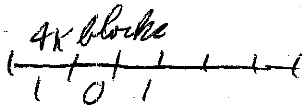
Interruption system [illegible] revised

Special reg.

etc

MCP & Multiprog

Bit-Map memory protection



Destruction-proof MCP

I/O channel

Disk information protection

2 strata interruption scheme

Dynamic address [illegible] relocation

2-level "maint scanner"

30-40 ns compat.

100 ns

5 ns

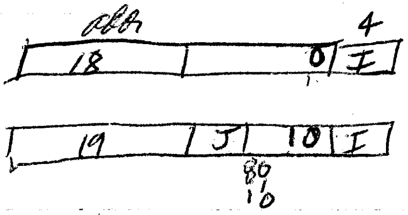
Index value field

no sign bit (also need sign on addr.)

progressive indexing -- hardware reduction

practically all instrs. are progressively indexed

F Pt.



Direct index instrs., LX, SL - halfwds, rest are full wds.

"Inter-index" instrs. XX, VV, etc

"Intra-index" instrs., shift, etc

Floating Pt.

3 accumulators

½ wds instr. incls

logical organ too

fraction overflow into V flag bit ← present problem in STRETCH

2X speed gain

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checking logical orders. can't do others {NOT / AND / OR / $\forall$}

①

A	A2
B	B2
C	C2

do ones in opposite order

② adder (reading out logic)

check this → ?

③ $P(A \cdot B) \forall P(A \vee B) = P(A \forall B) = P(A) \forall P(B)$

can use parity, too!

	True for	
$P(A) = P(\bar{A})$	even even	odd odd
$P(A) = \overline{P(\bar{A})}$	even odd	odd even

(FP) class I.

(18) YYX(XX)(XX)1$\overset{P}{X}$(I)

(0D)(00)
(01)(01)
(1D)(10) ← addr of accumulators.
11 111 ← return to self

class II
YYXX11(1)
ST
STN
N+
M−

class III
YYXX1111
mpy+add

(N) 000+
(U) 001−
010×
011/
100R−

0101 X+ Xor
1101 X− Inter
and
or

VFL "flow mode" 1.2μs, 1st byte 0.6μs subsequent

Execute
Execute even if Br,
LVE1 no index addr.